

Malawi Food Security Monitor

Food Price Spike Analysis & District Risk Assessment

Analysis Period: January 2020 – April 2026
Data Source: World Food Programme (WFP) VAM Food Price Database
Geographic Scope: 28 Districts · 113 Markets · 3 Regions
Report Generated: 12 May 2026
Classification: For Official Use — NGO / Government Partners

Executive Summary

Key findings from the automated food price spike detection system

MARKETS MONITORED 113 28 districts covered	CRITICAL SPIKES 215 of 2,595 total events	HIGHEST RISK DISTRICT Machinga Score: 436.0	MONITORING GAPS 5 critical gap districts
---	--	--	---

Key Finding: Machinga District in Southern Region has the highest food security risk score (436.0) in Malawi. 215 critical price spike events were detected across the analysis period. The most affected commodity is **Maize** with 123 critical events — representing over 70% of all critical spikes.

Methodology: Spike detection uses a dual-method approach requiring both month-over-month percentage change $\geq 20\%$ AND rolling 12-month z-score ≥ 1.5 . This eliminates false positives from seasonal variation. Severity follows the WFP ALPS (Alert for Price Spikes) framework.

SEVERITY LEVEL	THRESHOLD	COUNT	% OF OBSERVATIONS	RECOMMENDED ACTION
Critical	$\geq 60\%$ change + z-score ≥ 2.5	215	8.29%	Immediate intervention required
Severe	$\geq 40\%$ change + z-score ≥ 2.0	799	30.79%	Prepare emergency response
Moderate		1581	60.92%	Monitor closely

	≥20% change + z-score ≥1.5			
Normal	Below thresholds	0	—	No action required

District Risk Rankings

All 28 districts ranked by weighted composite risk score (Critical×3 + Severe×2 + Moderate×1)

#	DISTRICT	REGION	RISK LEVEL	RISK SCORE	CRITICAL	SEVERE	MODERATE	SPIKE RATE
1	Machinga	Southern Region	Critical	436	25	79	203	0.0%
2	Mulanje	Southern Region	Critical	277	13	65	108	0.0%
3	Zomba	Southern Region	High	262	21	54	91	0.0%
4	Nsanje	Southern Region	High	252	11	41	137	0.0%
5	Chikwawa	Southern Region	High	217	9	47	96	0.0%
6	Dowa	Central Region	High	199	9	34	104	0.0%
7	Ntcheu	Central Region	Moderate	185	9	45	68	0.0%
8	Ntchisi	Central Region	Moderate	175	11	36	70	0.0%
9	Phalombe	Southern Region	Moderate	167	8	37	69	0.0%
10	Thyolo	Southern Region	Moderate	156	5	40	61	0.0%
11	Balaka	Southern Region	Moderate	126	9	21	57	0.0%
12	Blantyre	Southern Region	Low	125	7	26	52	0.0%
13	Mangochi	Southern Region	Low	119	4	30	47	0.0%
14	Mzimba	Northern Region	Low	110	9	22	39	0.0%
15	Mchinji	Central Region	Low	108	6	27	36	0.0%
16	Karonga	Northern Region	Low	104	9	21	35	0.0%
17	Rumphi	Northern Region	Low	92	6	21	32	0.0%
18	Kasungu	Central Region	Low	89	5	21	32	0.0%
19	Chitipa	Northern Region	Low	87	7	20	26	0.0%
20	Salima	Central Region	Low	87	6	18	33	0.0%
21	Dedza	Central Region	Low	78	3	15	39	0.0%
22	Lilongwe	Central Region	Low	65	6	14	19	0.0%

23	Nkhata Bay	Northern Region	Low	59	4	8	31	0.0%
24	Neno	Southern Region	Stable	53	4	16	9	0.0%
25	Mwanza	Southern Region	Stable	48	4	11	14	0.0%
26	Nkhotakota	Central Region	Stable	43	3	9	16	0.0%
27	Chiradzulu	Southern Region	Stable	14	1	1	9	0.0%
28	Likoma	Northern Region	Stable	0	0	0	0	0.0%

Critical Spike Events

Most recent critical price spike events (price jump ≥60% + z-score ≥2.5)

DATE	DISTRICT	MARKET	COMMODITY	PRICE (MWK)	% JUMP
2026-03-15	Mulanje	Chinakanaka	Sweet potatoes	2,778 MWK	+399.6%
2026-02-15	Blantyre	Lunzu	Groundnuts (shelled)	16,000 MWK	+77.8%
2025-11-15	Salima	Kamuzu Road	Sorghum	3,500 MWK	+250.0%
2025-10-15	Blantyre	Lirangwe	Sorghum	5,473 MWK	+376.0%
2025-10-15	Machinga	Nayuchi	Millet	3,331 MWK	+375.9%
2025-10-15	Blantyre	Lunzu	Millet	8,333 MWK	+733.3%
2025-10-15	Rumphi	Rumphi Boma	Pigeon peas	4,500 MWK	+150.0%
2025-10-15	Phalombe	Chilinga	Cowpeas	5,000 MWK	+66.7%
2025-09-15	Lilongwe	Area 23	Maize	1,222 MWK	+619.0%
2025-09-15	Chikwawa	Tomali	Maize	1,547 MWK	+336.0%
2025-09-15	Machinga	Chikweo	Maize	1,586 MWK	+373.6%
2025-09-15	Karonga	Karonga Boma	Maize	1,150 MWK	+232.1%
2025-09-15	Chikwawa	Nkhate	Maize	1,300 MWK	+271.4%
2025-09-15	Chitipa	Nthalire	Maize	1,001 MWK	+203.5%
2025-09-15	Chitipa	Chitipa Boma	Maize	989 MWK	+199.9%
2025-09-15	Karonga	Uliwa	Cowpeas	4,500 MWK	+155.2%
2025-09-15	Chikwawa	Dyelatu	Maize	1,573 MWK	+362.9%
2025-09-15	Chitipa	Kameme	Maize	956 MWK	+184.3%
2025-09-15	Blantyre	Lirangwe	Maize	1,556 MWK	+357.8%
2025-09-15	Dowa	Dowa Boma	Maize	1,275 MWK	+325.0%

Commodity Analysis

Which commodities experienced the most price shocks

COMMODITY	CRITICAL EVENTS	TOTAL SPIKES	SPIKE RATE
Maize	123	233	9.0%
Maize (new harvest)	31	282	10.9%
Pigeon peas	18	299	11.5%
Cowpeas	16	371	14.3%
Beans	13	395	15.2%
Rice	4	82	3.2%
Sweet potatoes	2	184	7.1%
Sorghum	2	68	2.6%
Millet	2	93	3.6%
Cassava	1	68	2.6%
Oil (vegetable)	1	82	3.2%
Sugar	1	137	5.3%
Groundnuts (shelled)	1	55	2.1%
Local vegetables	0	55	2.1%
Potatoes	0	3	0.1%

Market Monitoring Gaps

Districts with high food security risk but insufficient market monitoring coverage

Why this matters: A district with high risk score but few monitored markets means food crises may go undetected. These gaps represent priority areas for WFP monitoring expansion.

DISTRICT	REGION	RISK SCORE	MARKETS	MONITORING STATUS
Machinga	Southern Region	436	8	CRITICAL GAP
Mulanje	Southern Region	277	5	CRITICAL GAP
Zomba	Southern Region	262	5	CRITICAL GAP
Nsanje	Southern Region	252	6	CRITICAL GAP
Chikwawa	Southern Region	217	5	CRITICAL GAP
Dowa	Central Region	199	7	ADEQUATE
Ntcheu	Central Region	185	5	ADEQUATE
Ntchisi	Central Region	175	5	ADEQUATE
Phalombe	Southern Region	167	5	ADEQUATE
Thyolo	Southern Region	156	5	ADEQUATE
Balaka	Southern Region	126	5	ADEQUATE
Blantyre	Southern Region	125	4	ADEQUATE
Mangochi	Southern Region	119	7	ADEQUATE
Mzimba	Northern Region	110	7	ADEQUATE
Mchinji	Central Region	108	3	HIGH GAP
Karonga	Northern Region	104	2	HIGH GAP
Rumphi	Northern Region	92	2	ADEQUATE
Kasungu	Central Region	89	3	ADEQUATE
Chitipa	Northern Region	87	3	ADEQUATE
Salima	Central Region	87	3	ADEQUATE
Dedza	Central Region	78	4	ADEQUATE
Lilongwe	Central Region	65	5	ADEQUATE
Nkhata Bay	Northern Region	59	3	ADEQUATE
Neno	Southern Region	53	1	MODERATE GAP
Mwanza	Southern Region	48	1	ADEQUATE
Nkhotakota	Central Region	43	3	ADEQUATE
Chiradzulu	Southern Region	14	1	ADEQUATE
Likoma	Northern Region	0	0	NO MONITORING

